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climasus4r: An R Package for Reproducible Climate-Health Data Integration and Epidemiological Analysis

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Software

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Summary

`climasus4r` (v.1.0.0) is an R package for reproducible climate-health data integration and epidemiological analysis in Brazil. It enables researchers to combine health records with weather, satellite, and socioeconomic data that are otherwise scattered across incompatible government sources. The package provides workflows for importing, cleaning and linking Brazilian public health data with meteorological observations, gridded climate, air pollution, land-cover information and census data. The resulting datasets support a range of epidemiological designs and analytical strategies, including case-crossover and interrupted time-series designs, as well as distributed lag non-linear models (DLNMs), attributable-fraction and burden estimation, multi-city meta-analysis, and spatiotemporal risk modelling. Its 80+ functions operate on data structures enriched with a `climasus_df` S3 metadata object, with messages and outputs in Portuguese, Spanish and English for multilingual teams and public health practitioners in Brazil and Latin America.

Statement of Need

The need for tools that facilitate climate-health research is increasing as climate change affects the burden of climate-sensitive diseases worldwide (Romanello et al., 2021), including dengue risk from hydrometeorological hazards and urbanisation in Brazil (Lowe et al., 2021), heat-related mortality (Zhao et al., 2021), and projected temperature-related mortality under future climate scenarios (Gasparrini et al., 2017).

Brazil provides access to a rich set of public national and international datasets for climate-health research, including health records from the Brazilian Department of Health Informatics of the Unified Health System (DATASUS), weather-station data from the National Institute of Meteorology (INMET), census indicators from the Brazilian Institute of Geography and Statistics (IBGE), and gridded products such as ERA5-Land, MERRA-2, CAMS and CHIRPS. In practice, however, these datasets are difficult to integrate because they differ in file formats, spatial and temporal resolution, terminology, and data-access conventions. For example, DATASUS distributes microdata in a proprietary compressed format (.dbc), INMET

provides point-based observations at hourly, daily and monthly resolutions, and ERA5-Land provides gridded daily fields at an approximately spatial resolution of 9 km. Combining such sources requires reproducible decisions regarding spatial resolution, temporal aggregation, and exposure construction. Furthermore, research teams often repeat the same extraction, cleaning, harmonisation, and linkage steps in project-specific scripts, making analyses more difficult to reproduce and compare across studies.

The data-integration problem is particularly acute in environmental epidemiology, where records must be harmonised across systems, health outcomes linked to municipalities and socioeconomic indicators, and climate exposures aligned with health events under explicit lag assumptions. `climasus4r` addresses this gap through a unified R workflow that takes users from raw multi-source records to model-ready climate-health datasets. It targets Brazilian data infrastructure, but its overall structure is relevant to other settings in which health, climate and socioeconomic data are public but technically fragmented.

State of the field

Several R packages address parts of the climate-health workflow. `microdatasus` automates the download and preprocessing of DATASUS microdata but does not provide climate integration or epidemiological modelling (Saldanha et al., 2019). `datasus` focuses on pre-aggregated TABNET data (Prado Siqueira, 2024). `geobr` and `censobr` offer Brazilian geographic and census data, but do not link these data to climate-health exposures (Pereira & Barbosa, 2023; Pereira & Gonçalves, 2024). On the modelling side, `dlnm` and `mvmeta` offer tools for distributed-lag modelling and pooling, but assume that exposure and outcome data have already been prepared and linked (Gasparrini, 2011; Gasparrini & Armstrong, 2013). Recent global pipelines, such as the Python framework of Dasgupta et al. (2025), support broad multi-country data integration, but do not perform a workflow tailored to Brazilian health system data or DATASUS-based distributed-lag analyses (Dasgupta et al., 2025).

`climasus4r` links these layers within a single workflow: importing DATASUS records, standardising categorical variables, filtering by ICD-10 diagnosis and demographic profile, joining geographic and socioeconomic indicators, constructing aligned climate exposures; and passing the result to modelling functions. Each step is recorded in `climasus_df` metadata to support inspection, auditing and reproduction. Native Parquet/DuckDB support handles larger datasets, and Portuguese, Spanish and English output improves accessibility for multilingual teams.

Software design

Architecture and Key functions

Figure 1 summarises the architecture of `climasus4r`. Its 80+ `sus_*` functions are organized into three layers: data preparation, exposure and covariate integration, and epidemiological analysis and modelling. These layers are supported by diagnostic and visualisation routines and by an optional reproducible analytical pipeline for fully automated workflows.

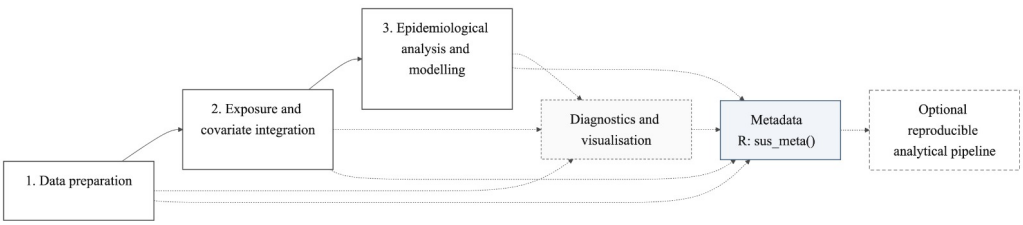


Figure 1: climasus4r climate-health workflow.

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The data preparation layer ingests records from DATASUS systems, Mortality (SIM), Hospital (SIH/SUS), Outpatient (SIA/SUS), Notifiable Diseases (SINAN), Health Establishments (CNES) and Live Births (SINASC) (Figure 2), importing (`sus_data_import()`), harmonising encodings/codes, defining case groups by ICD-10 and demographic filters, deriving variables, and aggregating into health time series (`sus_data*`).

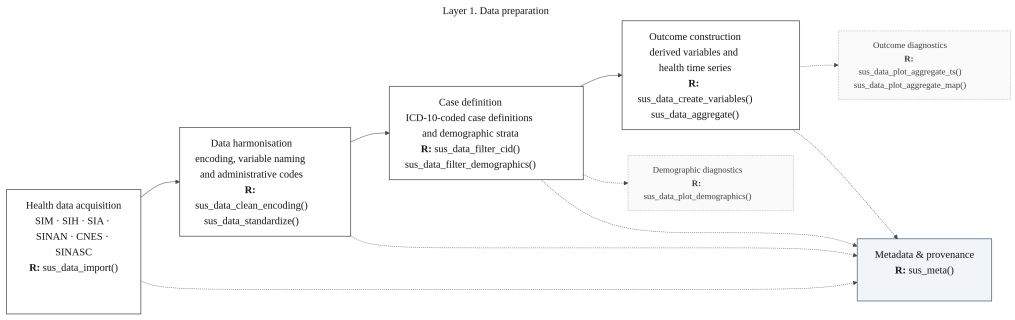


Figure 2: Package's data preparation layer.

The exposure and covariate integration layer links outcomes to geographic units and census indicators (`sus_spatial_join()`, `sus_census_join()`), gap-fills INMET station observations, integrates gridded products such as ERA5, CHIRPS, air pollution, drought and fire data (`sus_grid*`), and builds aligned exposure records for modelling (Figure 3).

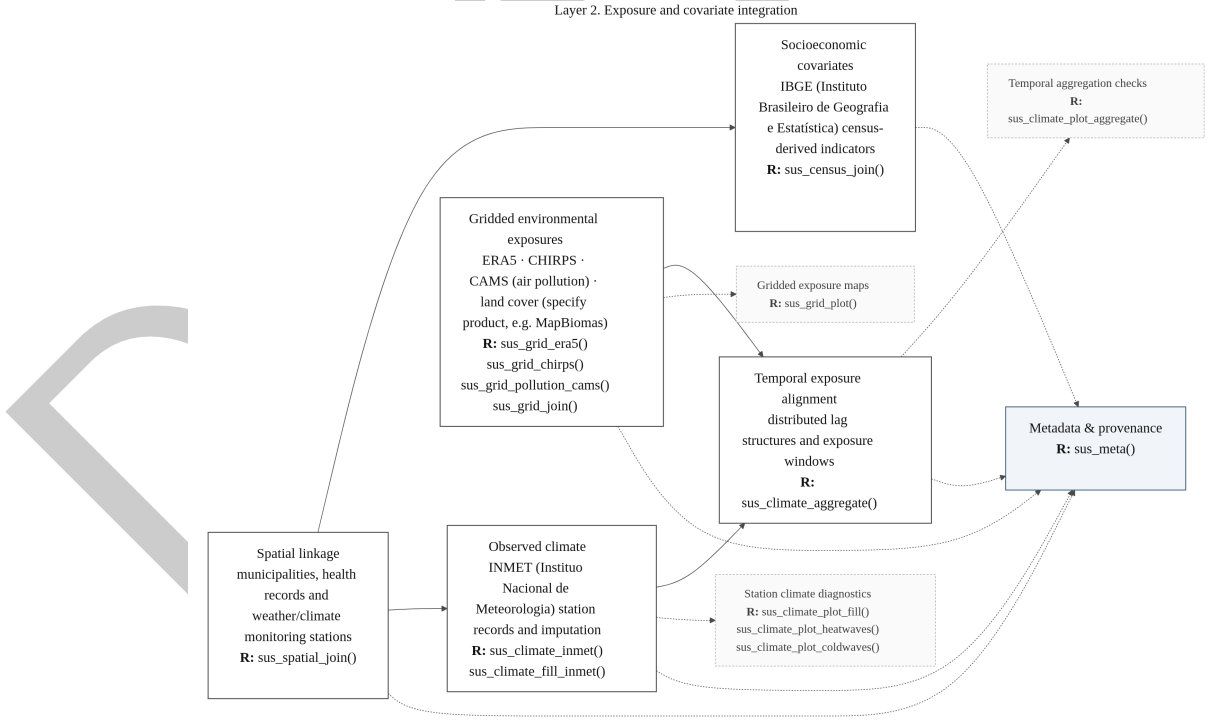


Figure 3: Package's exposure and covariant integration layer.

The epidemiological analysis and modelling layer supports non-linear models (`sus_mod_dlnm()`), case-crossover and interrupted time-series designs, attributable-fraction and burden estimation, multi-city pooling, and spatial risk analysis (`sus_mod*`), with companion plotting functions for model diagnostics (Figure 4).

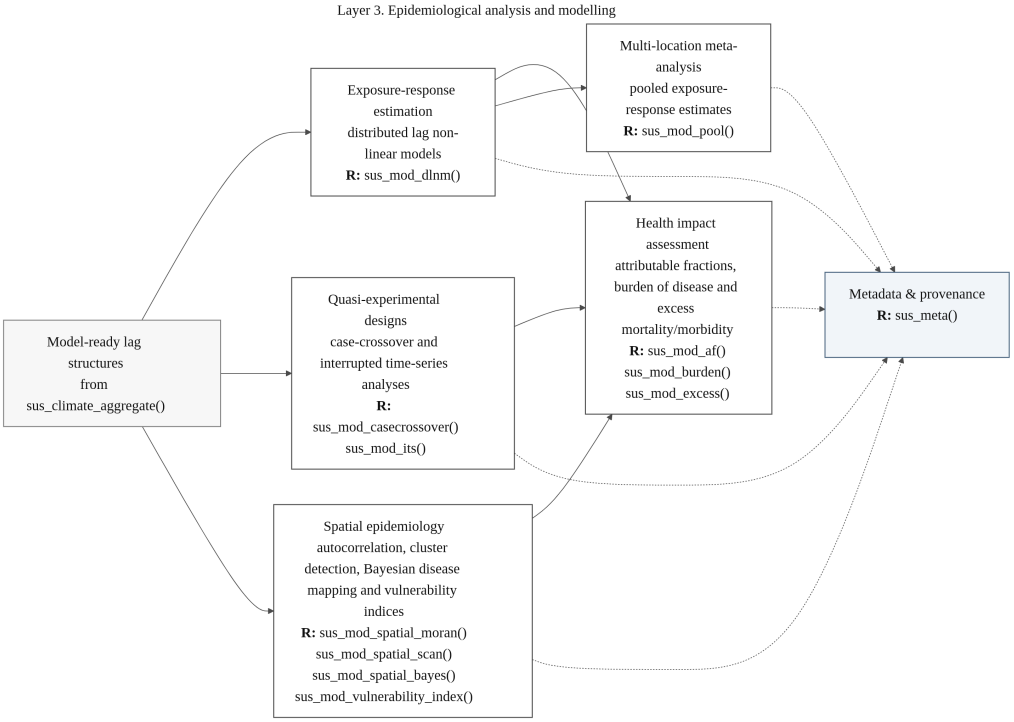


Figure 4: Package's epidemiological analysis and modelling layer.

Across all layers, `climasus4r` records data provenance in a `climasus_df` metadata object, accessed with `sus_meta()`, tracking sources, stages and transformation history in outputs compatible with standard R workflows (Figure 5). Recorded choices can be exported via `sus_rap_recipe()`, `sus_rap_targets()` and `sus_rap_export()` as a reproducible analytical pipeline, including `targets` support.

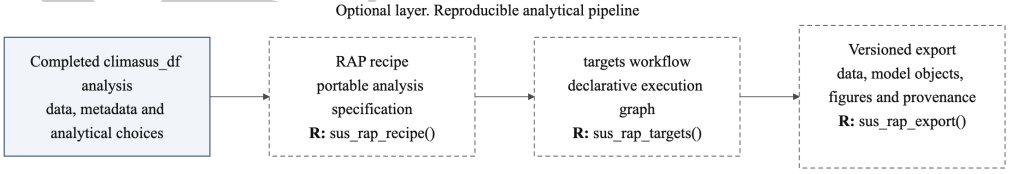


Figure 5: Package's metadata and reproducible analytical pipeline layer.

Temporal exposure-alignment strategies

The main exposure-construction function, `sus_climate_aggregate()`, converts raw climate series into lagged, windowed or threshold-based exposure variables required by statistical models used in epidemiological analyses. It exposes ten named strategies, each corresponding to a specific exposure-lag assumption (Figure 6). The selected strategy is stored in `climasus_df` metadata, allowing exposure construction to be audited alongside the model that uses it.

Each strategy is anchored to a published design rather than a generic smoothing choice: `exact/distributed_lag` follow same-day and distributed-lag mortality designs used in Brazilian cities (Bell et al., 2008; Gasparrini et al., 2010; Son et al., 2016); `moving_window/offset_window` match incubation-period assumptions in dengue climate-suitability models (Lee et al., 2021); `degree_days` operationalises thermal-development thresholds for *Aedes aegypti* (Grech et al., 2015); `seasonal` supports ecological associations such as leishmaniasis and biome-level climate variation (Kersul et al.,

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2025); `threshold_exceedance`/`cold_wave_exceedance` implement heat-wave and cold-spell definitions for cause-specific mortality in São Paulo (Geirinhas et al., 2021; Moraes et al., 2022); and `weighted_window` follows the harvesting-resistant weighted distributed-lag approach from air-pollution research (Bhaskaran et al., 2013; Zeger et al., 1999). Default thresholds reflect Brazilian tropical, subtropical and temperate regions and can be overridden by the user.

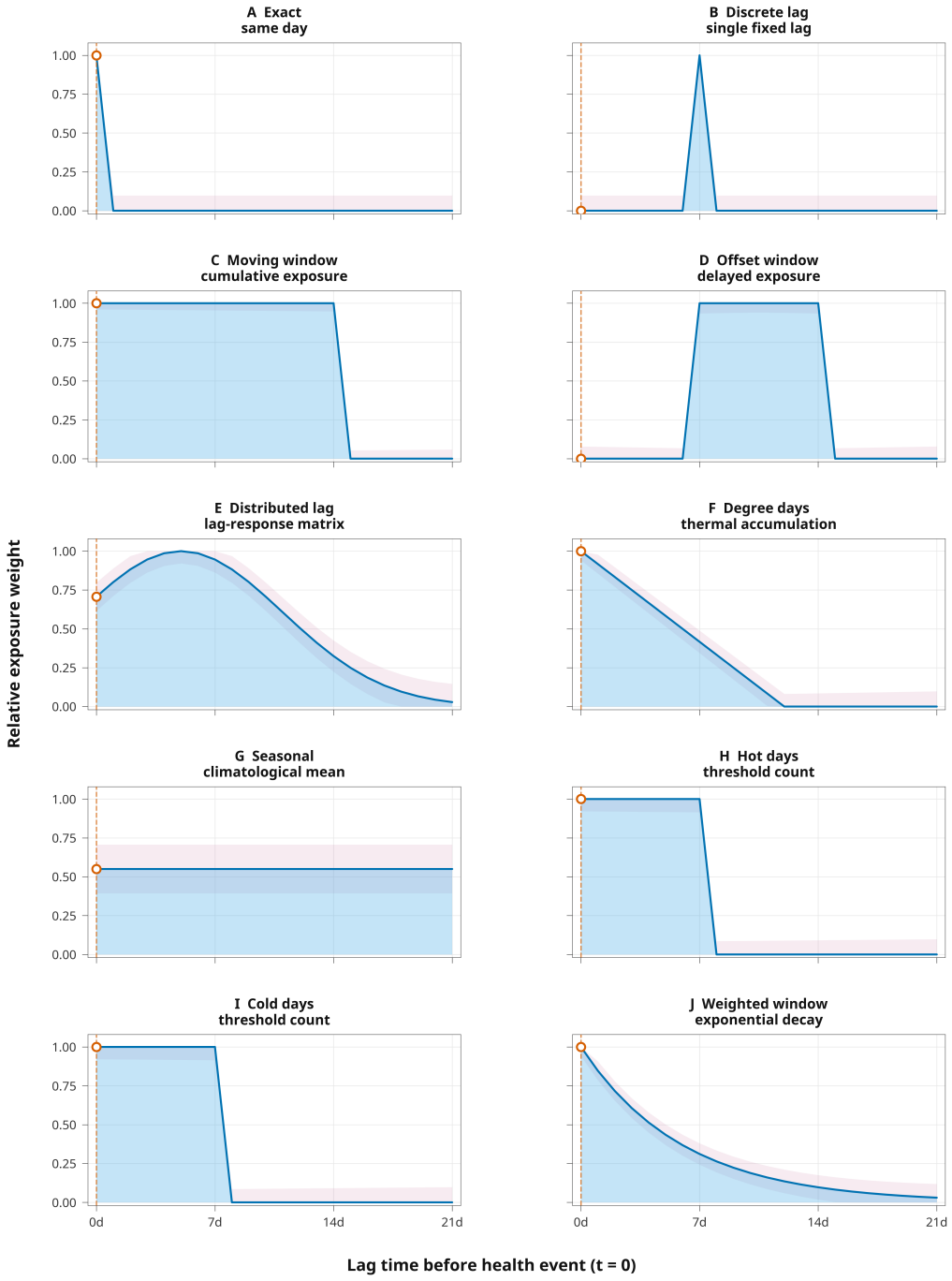


Figure 6: Temporal exposure-alignment strategies in `sus_climate_aggregate()`. Each panel shows the relative exposure weight by lag day before a health event (`t = 0`, dashed red line), spanning same-day, fixed-lag, cumulative, delayed, distributed, seasonal, threshold-based and decay-weighted windows. Weights are schematic, not fitted exposure-response functions.

Research impact statement

`climasus4r` was developed for and is in active use within the INCT-CONEXAO BIO3TOX project (CNPq Process 408474/2024-6), where it underpins the climate-health data-integration pipeline linking DATASUS records to meteorological and environmental exposures, and its continued development is supported by the PRÓ-AMAZÔNIA project (CNPq Process 442500/2025-4). Near-term community-readiness signals include a companion Python interface, `climasus4py`, and a no-code desktop pipeline, `climasus+ Studio`, both under active development to extend the same reproducible workflow to non-R users, and multilingual (Portuguese/Spanish/English) tutorials aimed at public health practitioners across Brazil and Latin America who are the intended adopters of the package.

The following section illustrates a complete `climasus4r` workflow as a reproducible, runnable example, from importing mortality records to fitting a distributed lag non-linear model, using respiratory deaths among children under five years in São Paulo state (2010-2019). It demonstrates the interoperability of objects across functions through the pipe operator and is a software walkthrough, not an epidemiological analysis. The example is restricted to one municipality and a narrow age stratum, and the ten-year period is chosen only to give the DLNM enough data to fit without error near the tails of the temperature distribution (Gasparrini et al., 2015). Users should adapt the study period, scale, strata and model specification to their own research question; the figures below (Figure 7, Figure 8 and Figure 9) illustrate what the package produces and should not be read as findings about the temperature-mortality relationship.

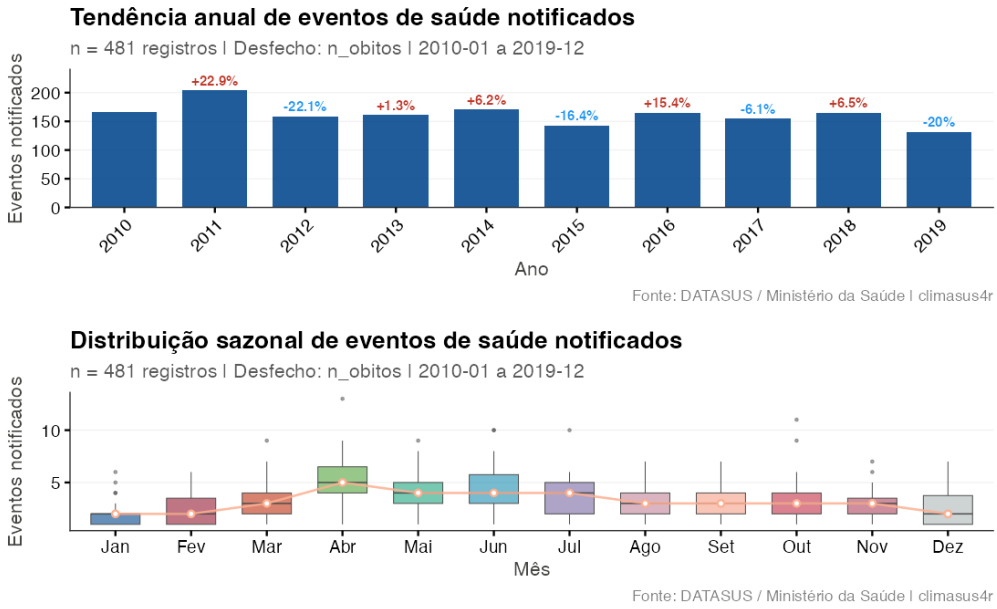


Figure 7: Trend and seasonal decomposition of weekly respiratory deaths in children under five, São Paulo city, 2010-2019 (log-transformed y-axis). Upper panel: smoothed ten-year trend. Lower panel: climatological seasonal pattern by calendar month.

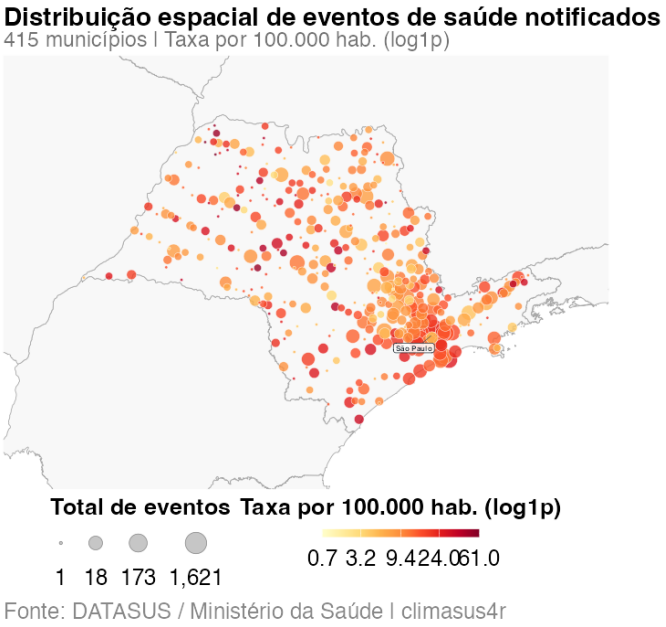


Figure 8: Incidence of respiratory deaths in children under five by municipality, São Paulo state, 2010-2019, per 100,000 children, as a bubble map with log-scaled colour and size, avoiding domination by São Paulo city's population size.

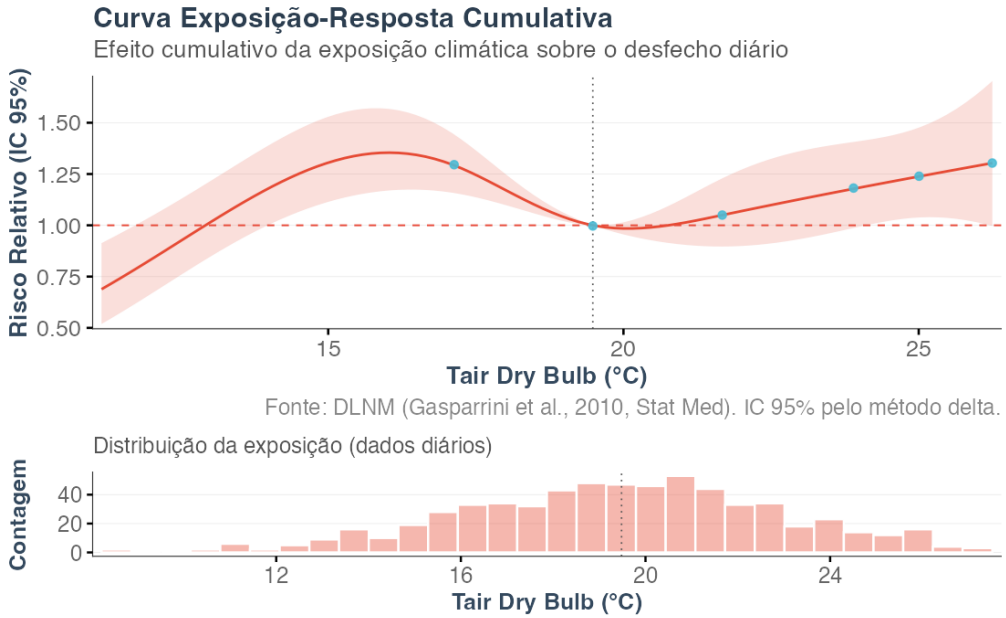


Figure 9: Example cumulative exposure-response output (DLNM) relating daily mean temperature to respiratory mortality in children under five, São Paulo city, 2010-2019, using a 21-day lag window (`sus_mod_dlnm()`). Band is the 95% CI and the histogram is the temperature distribution. The plot confirms that the `climasus_df` object from `sus_climate_aggregate(temporal_strategy = "distributed_lag")` passes directly to `sus_mod_dlnm()`/`sus_mod_plot_dlnm()`; given the narrow scope of the example, the shape of the curve illustrates software output only and should not be interpreted as evidence about the temperature-mortality relationship in São Paulo.

Tutorial and Ecosystem

Tutorials in Portuguese, English and Spanish are available at https://bymaxanjos.github.io/climasus4r/. They cover climate indicators, heat and cold events, applied climate-epidemiology examples, and reproducible analytical pipelines. `climasus4r` is part of an ecosystem that includes `climasus4py`, a Python interface for Python-based workflows (https://github.com/climasus), and climasus+ Studio, a no-code pipeline interface desktop (https://github.com/ByMaxAnjos/climasus-plus).

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AI usage disclosure

The authors used Claude Code (Anthropic), a large language model-based coding assistant, to support review and optimisation of selected R scripts, with attention to coding style, reproducibility and maintainability. All AI-assisted outputs were reviewed, edited and verified by the authors for technical accuracy and consistency with the aims of the manuscript and software. The authors take full responsibility for the final content, analyses and code reported in this publication.

Conflict of Interest

The authors declare no conflicts of interest.

References

Bell, M. L., O'Neill, M. S., Ranjit, N., Borja-Aburto, V. H., Cifuentes, L. A., & Gouveia, N. C. (2008). Vulnerability to heat-related mortality in latin america: A case-crossover study in são paulo, brazil, santiago, chile and mexico city, mexico. *International Journal of Epidemiology*, *37*(4), 796–804. https://doi.org/10.1093/ije/dyn094

Bhaskaran, K., Gasparrini, A., Hajat, S., Smeeth, L., & Armstrong, B. (2013). Time series regression studies in environmental epidemiology. *International Journal of Epidemiology*, *42*(4), 1187–1195. https://doi.org/10.1093/ije/dyt092

Dasgupta, A., Perez-Fernandez, I., Huynh, T., Mills, C., Nicholls, R. C., Sambaturu, P., Choisy, M., Wallom, D., Nguyen-Duy, T., Inward, R. P. D., Brittain, J.-S., Sparrow, S., & Kraemer, M. U. G. (2025). Scalable, open-access and multidisciplinary data integration pipeline for climate-sensitive diseases. *Wellcome Open Research*, *10*, 467. https://doi.org/10.12688/wellcomeopenres.24774.2

Gasparrini, A. (2011). Distributed lag linear and non-linear models in R: The package dlnm. *Journal of Statistical Software*, *43*(8), 1–20. https://doi.org/10.18637/jss.v043.i08

Gasparrini, A., & Armstrong, B. (2013). Reducing and meta-analysing estimates from distributed lag non-linear models. *BMC Medical Research Methodology*, *13*, 1. https://doi.org/10.1186/1471-2288-13-1

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Gasparrini, A., Armstrong, B., & Kenward, M. G. (2010). Distributed lag non-linear models. *Statistics in Medicine*, *29*(21), 2224–2234. https://doi.org/10.1002/sim.3940

Gasparrini, A., Guo, Y., Hashizume, M., Lavigne, E., Zanobetti, A., Schwartz, J., Tobias, A., Tong, S., Rocklov, J., Forsberg, B., Leone, M., De Sario, M., Bell, M. L., Guo, Y.-L. L., Wu, C., Kan, H., Yi, S.-M., Sousa Zanotti Stagliorio Coelho, M. de, Saldiva, P. H. N., … Armstrong, B. (2015). Mortality risk attributable to high and low ambient temperature: A multicountry observational study. *The Lancet*, *386*(9991), 369–375. https://doi.org/10.1016/S0140-6736(14)62114-0

Gasparrini, A., Guo, Y., Sera, F., & others. (2017). Projections of temperature-related excess mortality under climate change scenarios. *The Lancet Planetary Health*, *1*(9), e360–e367. https://doi.org/10.1016/S2542-5196(17)30156-0

Geirinhas, J. L., Russo, A., Libonati, R., Sousa, P. M., Miralles, D. G., & Trigo, R. M. (2021). Recent increasing frequency of compound summer drought and heatwaves in southeast brazil. *Environmental Research Letters*, *16*(3), 034034. https://doi.org/10.1088/1748-9326/abe0eb

Grech, M. G., Sartor, P. D., Almirón, W. R., & Ludueña-Almeida, F. F. (2015). Effect of temperature on life history traits during immature development of Aedes aegypti and Culex quinquefasciatus (Diptera: Culicidae) from Córdoba city, Argentina. *Acta Tropica*, *146*, 1–6. https://doi.org/10.1016/j.actatropica.2015.02.010

Kersul, M. G., Donato, L. E., Santos, A. G. dos, Albuquerque e Silva, R., Sousa-Gomes, M. L. de, & Galvis-Ovallos, F. (2025). Climatic, environmental, and social factors in visceral leishmaniasis: A spatio-temporal perspective in Brazilian biomes. *PLOS Neglected Tropical Diseases*, *19*(12), e0013842. https://doi.org/10.1371/journal.pntd.0013842

Lee, S. A., Economou, T., Castro Catao, R. de, Barcellos, C., & Lowe, R. (2021). The impact of climate suitability, urbanisation, and connectivity on the expansion of dengue in 21st century brazil. *PLOS Neglected Tropical Diseases*, *15*(12), e0009773. https://doi.org/10.1371/journal.pntd.0009773

Lowe, R., Lee, S., O'Reilly, K. M., Brady, O. J., Bastos, L., Carrasco-Escobar, G., Castro Catão, R. de, Colón-González, F. J., Barcellos, C., Sá Carvalho, M., Blangiardo, M., Rue, H., & Gasparrini, A. (2021). Combined effects of hydrometeorological hazards and urbanisation on dengue risk in Brazil: A spatiotemporal modelling study. *The Lancet Planetary Health*, *5*(4), e209–e219. https://doi.org/10.1016/S2542-5196(20)30292-8

Moraes, S. L. de, Almendra, R., & Barrozo, L. V. (2022). Impact of heat waves and cold spells on cause-specific mortality in the city of São Paulo, Brazil. *International Journal of Hygiene and Environmental Health*, *239*, 113861. https://doi.org/10.1016/j.ijheh.2021.113861

Pereira, R. H. M., & Barbosa, R. J. (2023). *Censobr: Download data from Brazil's population census*. https://github.com/ipeaGIT/censobr

Pereira, R. H. M., & Gonçalves, C. N. (2024). *Geobr: Download official spatial data sets of Brazil*. https://doi.org/10.32614/CRAN.package.geobr

Prado Siqueira, R. (2024). *Datasus: An interface to DATASUS system*. https://github.com/rpradosiqueira/datasus

Romanello, M., McGushin, A., Di Napoli, C., Drummond, P., Hughes, N., Jamart, L., & others. (2021). The 2021 report of the Lancet Countdown on health and climate change: Code red for a healthy future. *The Lancet*, *398*(10311), 1619–1662. https://doi.org/10.1016/S0140-6736(21)01787-6

Saldanha, R. de F., Bastos, R. R., & Barcellos, C. (2019). *Microdatasus: A package for downloading and preprocessing microdata from the brazilian health informatics department (DATASUS)*. https://doi.org/10.1590/0102-311x00032419

DRAFT

Son, J.-Y., Gouveia, N., Bravo, M. A., Freitas, C. U. de, & Bell, M. L. (2016). The impact of temperature on mortality in a subtropical city: Effects of cold, heat, and heat waves in são paulo, brazil. *International Journal of Biometeorology*, *60*(1), 113–121. https://doi.org/10.1007/s00484-015-1009-7

Zeger, S. L., Dominici, F., & Samet, J. (1999). Harvesting-resistant estimates of air pollution effects on mortality. *Epidemiology*, *10*(2), 171–175. https://doi.org/10.1097/00001648-199903000-00015

Zhao, Q., Guo, Y., Ye, T., Gasparrini, A., Tong, S., Overcenco, A., Urban, A., Schneider, A., Entezari, A., Vicedo-Cabrera, A. M., & others. (2021). Global, regional, and national burden of mortality associated with non-optimal ambient temperatures from 2000 to 2019: A three-stage modelling study. *The Lancet Planetary Health*, *5*(7), e415–e425. https://doi.org/10.1016/S2542-5196(21)00081-4

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